

Question 3

Scenario:

1. Inside Packet Tracer:

- The layout of the network in packet tracer. (10)
- Devices can communicate with other devices within the network. (10)
- Labelling and naming conventions (10)
- Departments can connect to the DNS server (20)

2. Subnet plan:

Show all subnet addresses

Subnet 1 – 192.168.1.0 - 192.168.1.63

Subnet 2 - 192.168.1.64 - 192.168.1.127

Subnet 3 - 192.168.1.128 - 192.168.1.191

Subnet 4 - 192.168.1.192 - 192.168.1.255

First host IP addresses

Subnet 1 – 192.168.1.1

Subnet 2 - 192.168.1.65

Subnet 3 - 192.168.1.129

Subnet 4 - 192.168.1.193

Last host IP address according to the subnet

Subnet 1 – 192.168.1.62

Subnet 2 - 192.168.1.126

Subnet 3 - 192.168.1.190

Subnet 4 - 192.168.1.254

The broadcast according to each subnet.

Subnet 1 – 192.168.1.63

Subnet 2 - 192.168.1.127

Subnet 3 - 192.168.1.191

Subnet 4 - 192.168.1.255

The subnet mask the network would use

255.255.255.192 or /26

How you have allocated the subnet mask.

From /24 subdividing 2 more times gives us 4 subnets with 64 addresses and this is at subnet mask /26.

all the steps for setting up and configuring the Multilayer switch as well as the other switches within the organization with all the

I added each departments computers to their own vlan and then used an inter-vlan to allow communication between the vlans. More commands below

commands used to do the configurations.

This is the command used for the switches to add them to the vlan, with respective fastethernet numbers applied.

Enable

```
Conf t
Vlan 4
Int fa0/1
Switchport mode access
Switchport access vlan 4
```

This is the command used for the switches to link them to the multilayer switch.

```
Int fa0/6
Switchport mode trunk
```

This is the command added to the multilayer switch to setup vlans.

```
Ip routing
Int vlan 1
ip add 192.168.1.62 255.255.255.192
No sh
```

3. **Creating the DNS server each department will need to have its own website which each department can reach.**

- IT department (www.MITec.co.za)
- Accounts department (www.MAccounts.co.za)
- Delivery department (www.MDelivery.co.za)
- Transfer department (www.MTransfer.co.za)

4. **Floorplan of all departments and implemented devices** (5)

